

ThemeCloner: Holdings-Free Discovery of Thematic Exposure via Latent Factor Projection

Working Draft

Abstract

Thematic investing groups companies by exposure to a structural economic driver, such as artificial intelligence, clean energy, or cybersecurity, rather than by traditional sector or industry classification. The baskets used to define or express a theme in a portfolio are typically built manually, inherently focused on large cap companies given the availability of analyst coverage, and backward looking. It is rarely clear in advance which themes are economically coherent enough to trade. In this paper, we present ThemeCloner, an engine that does not use holdings and that extracts a theme's statistical signature from the comovement of thematic reference baskets, such as exchange traded funds (ETFs), in a broad large and midcap universe, and projects that signature onto a separate, unexplored universe of assets, such as a small-cap universe, to discover stocks with genuine thematic exposure that no classification or ETF holding has labeled.

Using principal component analysis on factor-residualized, standardized returns, our central result is a diagnostic: a theme produces clean, investable baskets if and only if it uniquely owns a latent factor. Concentration alone is not enough: a theme can be concentrated on a factor it shares with others, yielding baskets that look clean but are contaminated, so it is distinctiveness of factor ownership, not coherence per se, that determines tradeability. The same criterion also flags theme overlap and tests whether the discovered signals survive controls for stock level momentum. This gives a direct answer to the claim that thematic investing is simply momentum. To validate that the diagnostic measures genuine structure rather than rediscovering taxonomy, we calibrate it against GICS sector and subsector classifications as control labels. The calibration yields a striking result: sectors do not define the coherence ceiling. Coherence is regime dependent rather than taxonomic. In the 2018-26 sample, several cross sector themes are more distinctly owned latent factors than the GICS sectors they span, because a structural driver has fractured those sectors internally. The work offers both a portable discovery tool and a pre-trade test for when a theme is economically real rather than a marketing label.

1. Introduction

Thematic investing has become one of the most prominent organizing ideas in modern asset management. Investors increasingly want exposure to structural shifts such as artificial intelligence, the energy transition, cybersecurity, and automation, which cut across traditional sector taxonomies. The industry packages these ideas into thematic baskets and products, such as thematic ETFs, index provider theme indices, and sellside baskets. These group companies by shared exposure to a trend rather than by predefined sector classifications, such as the Global Industry Classification Standard (GICS) or the Industry Classification Benchmark (ICB). The premise is that the relevant axis of comovement is the theme, not the sector.

Three difficulties undermine this premise in practice. First, thematic baskets are biased toward large capitalization companies and are defined by holdings. They are built from the constituent lists of thematic products, so smaller companies that may carry a theme's purest exposure, but

that no reference basket includes, are invisible to them. Second, the baskets are manually constructed from holdings data and fundamental classification, which makes them costly to build, opaque, and difficult to extend to a universe the index provider has not already labeled. Third, there is no principled way to know in advance which themes are economically coherent enough to trade. Some themes correspond to genuine, distinct patterns of comovement, while others are taxonomic labels imposed on companies that do not actually move together. A recurring skeptical claim, voiced informally across the industry but never, to our knowledge, tested directly, sharpens this point: thematic investing may simply be momentum in disguise. Under this view, thematic baskets earn returns because their constituents have recently risen, not because they share a durable economic driver.

This paper addresses all three difficulties. We ask whether a theme's signature can be recovered from return comovement alone, without holdings data, and whether that signature can be projected onto a separate universe to discover thematically exposed stocks that no existing classification has labeled. We then ask what distinguishes a theme that yields clean, investable baskets from one that does not. Finally, we ask whether the discovered signal is genuinely distinct from momentum.

Our approach, ThemeCloner, treats thematic ETF returns as labels for a latent factor rather than as lists of constituents. We residualize a broad large and mid capitalization universe against standard risk factors, extract latent factors from the residual comovement using principal component analysis, and identify each theme's fingerprint by regressing its ETFs onto those factors. We then project a separate small capitalization universe into the same factor space and score each stock by its similarity to each fingerprint, yielding candidate baskets. The method never uses ETF holdings. The ETF return series is the theme definition.

Our central finding is diagnostic: a theme produces clean, investable baskets if and only if it uniquely owns a latent factor. Coherence, meaning how concentrated a theme's fingerprint is, is necessary but not sufficient. A theme can be concentrated on a factor it shares with other themes, producing baskets that look clean but are contaminated. Distinctiveness, meaning whether the theme owns its dominant factor, is the operative criterion. This same mechanism explains three otherwise separate phenomena: which themes yield good baskets, why commodity themes contaminate one another, and which themes survive a momentum control. It is, in effect, a test before trading of whether a theme is economically real, computable before any basket is built.

To establish that this diagnostic measures genuine structure rather than merely rediscovering sector taxonomy, we calibrate it against GICS sectors and subsectors as control labels. The natural prior is that sectors should define the coherence ceiling, since a sector should own a factor in the PCA model almost by definition. The calibration validates the diagnostic and produces a striking result in its own right: sector labels do not define a universal coherence ceiling. Coherence depends on regime rather than taxonomy. In our 2018 to 2026 sample, several themes that cut across sectors are more distinct owners of latent factors than the broad GICS sectors they draw from, because the dominant structural driver of the period, capital expenditure on artificial intelligence, has split those sectors internally. A theme that spans multiple sectors can be a tighter covariance cluster than the sector itself. That is strong evidence the diagnostic captures real structure, not an artifact of taxonomy. We develop this result as a validation of the method rather than as a separate claim about thematic investing's legitimacy.

We make four contributions. We present an engine for thematic stock discovery that does not use holdings and can operate across universes. We introduce a coherence and distinctiveness diagnostic that determines, before any basket is built, whether a theme is a genuine covariance cluster or a taxonomic label. This is the paper's central contribution. We provide direct evidence on the thematic versus momentum question, and show that momentum-independence is not predicted by factor ownership: the theme with the most robust independent signal is a contaminated one, while the two clean factor owners show no independent signal. Finally, by calibrating the diagnostic against GICS controls, we document that coherence depends on regime, which validates that the method resolves structure finer than standard sector classification.

2. Related Work

2.1 Latent Factor Models of Returns

Our factor extraction step builds on the literature that estimates latent asset pricing factors statistically rather than specifying them in advance. The risk-premium PCA (RP-PCA) framework of Lettau and Pelger generalizes principal component analysis (PCA) by penalizing pricing errors across the cross section. This can reveal economically important low variance factors that ordinary PCA, which ranks components by variance alone, can bury beneath high variance noise. This makes RP-PCA a natural starting point for theme discovery. In the language of this literature, themes are weak factors: they explain only a modest share of residual variance and resemble the low variance but real structure that RP-PCA is designed to recover. As we show in Section 3.3, however, the specific signal RP-PCA uses to identify weak factors is one our discovery objective does not use. In our setting, the method therefore reduces to ordinary PCA applied to a carefully constructed residual covariance.

2.2 Text-Based and Classification-Free Company Grouping

A related literature derives company relationships from sources other than returns. Industry networks built from regulatory filing text produce peer groups that differ from fixed classifications such as SIC or GICS, and recent work learns company embeddings from disclosures that recover industry and similarity structure. A complementary approach derives themes from disclosure language rather than return comovement and tests whether theme membership based on language anticipates later ETF inclusion (Amini et al., working paper). These approaches share our goal of moving beyond fixed taxonomies, but they operate on text. ThemeCloner operates on return covariance. The two sources provide complementary signals for the same underlying question: what economically groups firms.

2.3 Thematic Investing and Its Coherence

The closest work to ours examines thematic investing from a risk perspective, defining a theme through a factor model and testing whether thematic baskets capture coherent common risk. That work tests the coherence of given baskets. We differ in objective and direction: we discover basket members across separate universes, and we use coherence not only to validate baskets but also to predict which themes are tradeable at all. Related practitioner work argues that companies within a theme should correlate with the theme's macro drivers while decorrelating from adjacent sectors. This is the fundamental data counterpart to the statistical separation we measure.

2.4 Momentum and the Thematic Critique

The claim that thematic investing is repackaged momentum has not, to our knowledge, been tested directly. The most relevant evidence is that momentum is not a standalone factor, but rather the time series autocorrelation of other factor returns, and that factor momentum subsumes much of individual stock momentum. This frames our distinction precisely. Momentum is a temporal property of factor returns, while a theme is a contemporaneous covariance structure across the cross section. The two are different mathematical objects. We therefore test their separation empirically rather than asserting it.

3. Methodology

ThemeCloner extracts the statistical signature of an investment theme from the comovement of thematic exchange-traded funds (ETFs), then projects that signature onto a separate, unexplored universe of stocks to identify names with genuine thematic exposure. The method is holdings-free by construction: ETF return series are used as theme labels, and no constituent data is required at any stage. This section describes the three-universe architecture, the residualization and factor-extraction steps, and the diagnostics that determine when a theme is recoverable.

3.1 Three-Universe Architecture

The pipeline separates three roles that are often conflated in thematic analysis:

- Theme definition universe: Each theme is represented by two to three US listed thematic ETFs. Their returns define the theme. Holdings are never used.
- Covariance universe: This is the large and mid capitalization cross section on which latent factors are estimated. We use US listed stocks above a USD 1 billion market capitalization floor, approximately 2,500 names, restricted to a single time zone to avoid foreign exchange and closing price timing artifacts.
- Target universe: This is the small capitalization universe in which candidates are sought. It consists of Nasdaq and NYSE names below the covariance universe floor, after metadata filters, approximately 2,100 names. This is where the method searches for previously unlabeled thematic exposure.

Keeping these universes distinct is essential. The covariance universe must be broad and clean enough for themes to appear as genuine latent factors. The target universe must be separate so that discovered names are genuinely new rather than reidentified ETF constituents.

A subtler overlap risk is that a thematic ETF may hold some stocks that also appear in the target universe. If so, overlap could mechanically raise the correlation between a discovered basket and the ETF used to validate it. Two features limit this concern. First, holdings are never used in discovery: the ETF enters only through its return series, so the model cannot select a stock because it is an ETF constituent. Second, the market capitalization split puts most reference ETFs, which are liquidity constrained and weighted toward large and mid capitalization stocks, in a different size regime from the below threshold target universe. Any effective ETF exposure to a discovered candidate should therefore be small. We treat a direct screen for holdings overlap, excluding any target stock held by a theme ETF, as a robustness check for future work.

3.2 Residualization

Raw returns are dominated by market, size, value, and momentum exposures. Running factor extraction on raw returns causes high beta or high momentum stocks to load on many themes, producing pervasive contamination. We therefore residualize all return series against the Fama French five factors, optionally augmented with momentum and commodity sector controls, before factor extraction. Section 4 describes these controls. The residual is the component of each stock's return that standard risk factors cannot explain.

Residuals are then winsorized at the 1st and 99th percentiles and standardized to unit volatility. This converts the subsequent decomposition from a covariance analysis into a correlation analysis, preventing any single high volatility stock from dominating the extracted factors. This standardization step proved decisive. Without it, factors were driven by a small number of volatile names. With it, each factor became an interpretable sector, style, or thematic axis.

3.3 Factor Extraction

We extract $K=15$ latent factors from the standardized residuals of the covariance universe. Although risk-premium PCA (RP-PCA) is a natural starting point for recovering weak factors such as themes, as discussed in Section 2.1, our objective makes ordinary principal component analysis (PCA) the appropriate specification.

The reason follows from the difference between two objectives. RP-PCA identifies weak factors by using information in mean returns. Its penalty rewards factors that help price the cross section, so a low variance factor that carries a risk premium can be pulled forward. Theme discovery is not a pricing question. We ask whether a theme corresponds to a distinct pattern of comovement, not whether it earns a premium. To isolate that comovement, we residualize returns against standard risk factors and work with residuals that are mean zero in the estimation sample. The information RP-PCA relies on, pricing error in the cross section of mean returns, is therefore removed by design. In this setting, the penalty term is uninformative, and RP-PCA reduces to PCA on the residual covariance. We confirm this empirically: theme detectability is invariant to the RP-PCA penalty weight across a range spanning two orders of magnitude.

This raises an immediate question. If themes are weak factors, does ordinary PCA risk burying them beneath high variance components, which is the problem RP-PCA was designed to solve? In our setting, residualization performs that role through a different mechanism. By projecting out the Fama French factors and the dominant market and style variation before the decomposition, we remove the high variance structure that would otherwise dominate the leading components. Weak theme factors are therefore less likely to be buried when PCA is applied to the residual structure. For factor pricing, the RP-PCA premium penalty carries information and can improve on PCA. For factor discovery, that information is not part of the object of interest. Residualization, rather than the pricing penalty, is what makes weak factors visible. PCA on standardized residuals is therefore the appropriate tool.

The choice of $K=15$ is justified by a factor count sweep reported in the Appendix. Each label's distinctiveness is essentially invariant to K over the tested range, except for one theme whose

dominant factor resolves only at $K=15$ or above. We therefore use $K=15$, the smallest factor count that resolves all themes without degrading the others.

3.4 Theme Fingerprints and Projection

Each theme's ETF returns are residualized using the same procedure and regressed on the K extracted factors, yielding a K dimensional loading vector: the theme's fingerprint in factor space. For each theme, the fingerprint is the centroid of its constituent ETFs' loading vectors, which reduces individual ETF construction noise. Every stock in the target universe is then projected into the same factor space and scored by cosine similarity between its loading vector and each theme fingerprint. The highest scoring names for each theme form the candidate basket.

3.5 Diagnostics: Coherence and Distinctiveness

Two diagnostics, computed before basket construction, assess whether a theme is recoverable. Coherence measures how concentrated a theme's fingerprint is across factors. It is computed as the average of three normalized concentration metrics: a concentration ratio, a Herfindahl index, and an entropy based effective factor count. Distinctiveness measures whether a theme uniquely owns its dominant factor or shares it with other themes.

The distinction between sharing a dominant factor and sharing a secondary factor is central. Themes can overlap on secondary axes without contaminating one another, because those axes do not drive the projection used to form the basket. Contamination occurs when multiple themes share the same dominant factor, since that factor drives scoring and therefore determines which stocks enter the candidate basket. Distinctiveness therefore measures overlap on the dominant axis specifically, not overlap in general.

The accompanying factor sharing map records, for each factor, which themes load on it most strongly. As shown in Section 4, distinctiveness, rather than coherence alone, is the operative criterion for producing clean baskets.

3.5.1 Robustness: Style-Factor Controls

As a further check on whether the extracted factors proxy for standard equity style exposure rather than genuine thematic structure, we regress five style benchmarks, growth (IUSG, VUG), value (VLUE, VTV), quality (QUAL), momentum (MTUM), and growth-at-a-reasonable-price (SPGP), against the extracted factors using the same procedure as the theme fingerprints. None of the five style benchmarks concentrates on any theme's dominant factor as strongly as the theme itself, with one partial exception on factor_1. IUSG and VUG (growth) load positively on factor_1 at R^2 of 0.120 and 0.146. VTV (value) loads negatively at R^2 of 0.371, the single highest loading any style benchmark records on any factor; VLUE (value) barely loads on factor_1 at all, at R^2 of 0.002. Sign inspection of the underlying regression weights explains the pattern: Water loads negatively on factor_1, alongside VTV and a cluster of utility-sector stocks, while AI Infrastructure and Robotics & AI load positively, alongside IUSG, VUG, and semiconductor names. The factor is better described as a bipolar growth-versus-defensive axis than as a single shared theme, and Water's and AI Infrastructure's common R^2 on it reflects opposite-sign exposure rather than overlapping economic content. VLUE and VTV, both

nominally value benchmarks, load very differently on this factor; the difference is consistent with VTV's heavier utility and financials weighting relative to VLUE's more diversified multi-factor construction, and illustrates that a single-ETF style proxy can be sensitive to index methodology. This is a limitation of the R^2 -based distinctiveness metric, which is sign-agnostic by construction: two themes can register a high sharing score on the same factor while sitting at economically opposite poles of it.

Style ETF	Style	R^2 on factor_1	Sign on factor_1
IUSG	Growth	0.120	+
VUG	Growth	0.146	+
VLUE	Value	0.002	~0
VTV	Value	0.371	-

R^2 and sign of five style-factor ETFs on factor_1, the axis shared by Water, AI Infrastructure, Robotics & AI, and SECTOR: US Technology. VTV's negative loading exceeds Water's own R^2 on this factor; sign, not magnitude, distinguishes the growth pole (IUSG, VUG, AI Infrastructure, Robotics & AI) from the defensive pole (VTV, Water).

3.6 Validation

Candidate baskets are validated out of sample, in the sense that target-universe stocks are held out of factor estimation, against the theme ETFs along two axes. The first is correlation: whether the equal weighted candidate basket comoves with the theme ETF benchmark. The second is beta decomposition: we regress the basket return on the ETF benchmark return to separate comovement, captured by correlation, from amplitude, captured by beta, and then recover annualized alpha. This two axis validation distinguishes baskets that track the theme at full amplitude from those with muted exposure. It also separates the effect of excluding mega capitalization stocks from genuine signal attenuation.

3.7 Data and Proxies

Several inputs are proxied because of data access constraints. Each proxy is noted for reproducibility. MSCI ACWI constituents require a vendor license and are proxied by free index lists, ultimately restricted to US listings. ETF constituent holdings are not used. ETF return series serve as theme labels, which is the method's holdings free feature rather than a limitation. The Russell 2000 constituent list requires an FTSE Russell license and is proxied by a metadata filtered Nasdaq and NYSE cross section. There is no canonical commodity factor library, so we proxy the Bakshi, Gao, and Rossi commodity factors with liquid commodity sector ETFs.

4. Results

We organize the results as a sequence of findings, each building on the previous one. We report results for eleven themes that cut across sectors: Robotics and AI, AI Infrastructure, Clean Energy, Cybersecurity, Defense, Fintech, US Infrastructure, Agribusiness, Critical Minerals, Water, and Timber and Forestry. We also report six calibration controls: three GICS sectors, Financials, Technology, and Energy, and three GICS subsectors, Banks, Software, and Biotech. Themes are admitted only if their defining ETF ensemble spans multiple GICS sectors. Single industry funds, such as biotech ETFs, are excluded as themes and used instead as subsector controls. This admissibility rule is applied before estimation, not selected after observing results.

4.1 Themes Are Weak but Recoverable Latent Factors

After residualization and standardization, each extracted factor corresponds to an interpretable sector, style, or thematic axis. Examples include utilities versus semiconductors, banks versus energy, and REITs versus homebuilders. Themes appear as latent factors with best single factor R^2 values of approximately 0.08 to 0.21. These are weak factors by construction: they explain only a modest share of residual variance. They are nevertheless clearly present and economically interpretable, rather than statistical artifacts.

[FIGURE PLACEHOLDER — Figure 2: theme x factor R^2 heatmap — outputs/fig_factor_interpretation.png]

4.2 Clean Factors Require a Clean Universe

This finding is methodological but consequential. Before imposing the market capitalization floor and standardizing residuals, extracted factors were dominated by a small number of highly volatile stocks. In one instance, a single stock carried a loading of 0.89. Penny stocks and crypto related names also distorted the factor structure. Two interventions were decisive: a USD 1 billion market capitalization floor for the covariance universe, and winsorization followed by unit volatility standardization of residuals. After both steps, the factor structure became cleanly interpretable. Statistical theme detection is only as reliable as the covariance universe on which the factors are estimated.

4.3 A Theme Is Tradeable If and Only If It Uniquely Owns a Factor

This is the central finding. Coherence alone does not predict basket quality. A theme can be highly concentrated yet still produce contaminated baskets if its dominant factor is shared. The operative criterion is distinctiveness: a theme produces clean, correct baskets when it uniquely owns a distinct latent factor, and suffers mutual contamination when several themes load most strongly on the same factor. The factor sharing map below, from the baseline specification, makes the mechanism explicit.

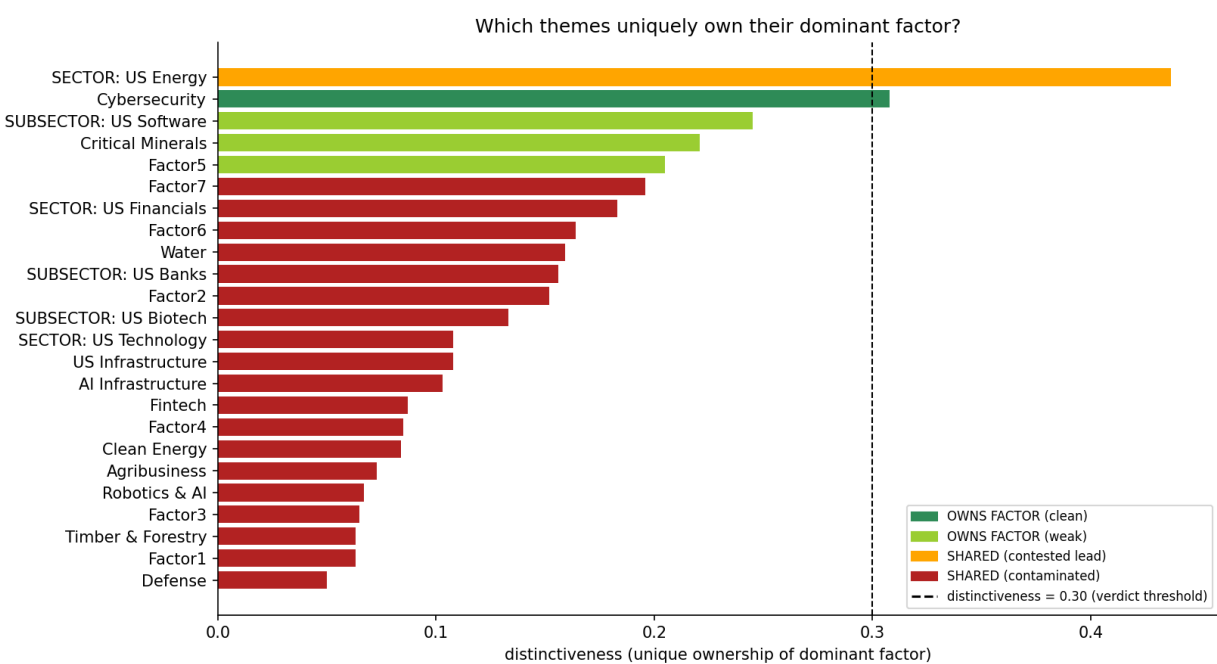
The distinction between sharing a dominant factor and sharing a secondary factor is central. Themes can overlap on secondary axes without contaminating one another, because those axes do not drive the projection used to form the basket. Contamination occurs when multiple themes share the same dominant factor, since that factor drives scoring and therefore determines which stocks enter the candidate basket. Distinctiveness therefore measures overlap on the dominant axis specifically, not overlap in general.

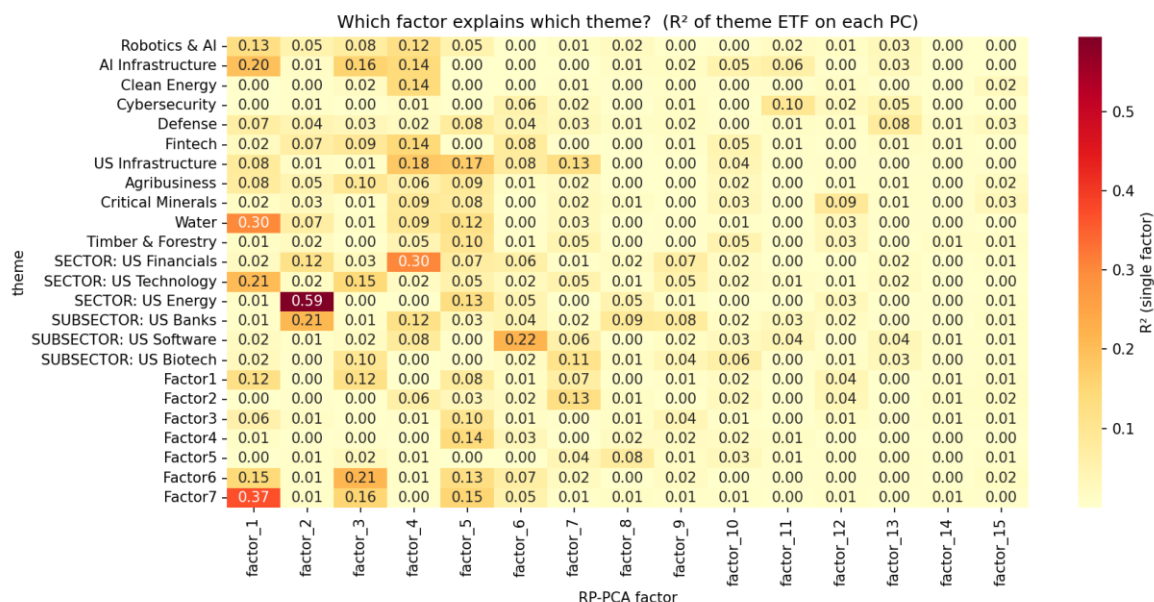
A worked example illustrates the mechanism. Cybersecurity, Critical Minerals, and Agribusiness each uniquely own a distinct factor despite loading partially on several common factors, which could otherwise suggest mutual contamination. Cybersecurity peaks on factor_11, Critical Minerals on factor_12, and Agribusiness, more weakly, on factor_3. Each theme is therefore identifiable and separable into its own basket despite secondary overlap. By contrast, four themes, Water, SECTOR: US Technology, AI Infrastructure, and Robotics & AI, all peak on the same factor_1, an axis that separates semiconductor and AI-hardware names from electric utilities. A second four-way pileup occurs on factor_4, shared by SECTOR: US Financials, US Infrastructure, Fintech, and Clean Energy, and a third, two-way pileup occurs on factor_5, shared by Timber & Forestry and Defense. Each of these themes is identifiable as a distinct axis of comovement but is not cleanly separable from the other labels sharing its factor, and its basket is correspondingly contaminated.

Factor	Themes loading on it (baseline)
factor_1	Water, SECTOR: US Technology, AI Infrastructure, Robotics & AI (shared)
factor_2	SECTOR: US Energy, SUBSECTOR: US Banks (shared)
factor_3	Agribusiness (unique, weak)
factor_4	SECTOR: US Financials, US Infrastructure, Fintech, Clean Energy (shared)
factor_5	Timber & Forestry, Defense (shared)
factor_6	SUBSECTOR: US Software (unique)
factor_7	SUBSECTOR: US Biotech (unique)
factor_11	Cybersecurity (unique, clean)
factor_12	Critical Minerals (unique, clean)

Table 1. Factor-sharing map, baseline specification (FF5 residualization, no commodity controls, K=15). Three themes uniquely own a factor and yield clean baskets: Cybersecurity (factor_11), Critical Minerals (factor_12), and, weakly, Agribusiness (factor_3). The remaining themes share their dominant factor with at least one other label and are correspondingly contaminated.

[FIGURE PLACEHOLDER — Figure 1: distinctiveness bar chart, baseline specification (17 labels) — outputs/fig_theme_coherence_baseline.png]





The themes that uniquely own a factor, Cybersecurity and Critical Minerals, and, more weakly, Agribusiness, produce the cleanest baskets. Themes that share factor_1, a semiconductor-versus-utilities axis, or factor_4, a broad cyclical and financials axis, bleed into one another instead. Clean Energy's basket is contaminated for a related but distinct reason, addressed directly in Section 4.4: its candidates load heavily on shared commodity exposure rather than on a genuine clean-energy signal.

4.4 Commodity-Linked Themes Fuse via Shared Commodity Beta

Clean Energy's basket contamination has an economic explanation distinct from its coherence-level factor sharing. Clean energy and mining companies share substantial commodity input and capital cycle exposure; even where the coherence diagnostic separates two themes onto different dominant factors, shared commodity exposure can still pull commodity-linked names, such as gold and lithium miners, into a basket built purely from return comovement. Adding commodity sector ETF controls to the residualization, as proxies for the Bakshi, Gao, and Rossi commodity factors, breaks this apart; after commodity residualization the affected baskets improve materially. We treat this as an extension rather than part of the baseline specification used elsewhere in this paper, since it requires an additional, theme-specific modeling choice rather than a single universal control.

Theme	Baseline basket character	With commodity controls
Clean Energy	Gold miners (AU, GFI, AEM, KGC, WPM, FNV)	Genuine clean-energy (BELFA, ENS, WATT, ASPN, CSIQ, TAC)
Agribusiness	Pipelines and miners (ENB, FNV, KGC)	Food and agriculture (POST, PPC, BJ, GEF)

Table 2. Effect of commodity-factor controls on two contaminated themes. Clean Energy ETF correlation rose from 0.70 to 0.88 after commodity residualization.

This is a limitation and a prescription in one. Themes whose narrative driver overlaps with a commodity factor require explicit commodity controls to separate the thematic signal from shared commodity beta.

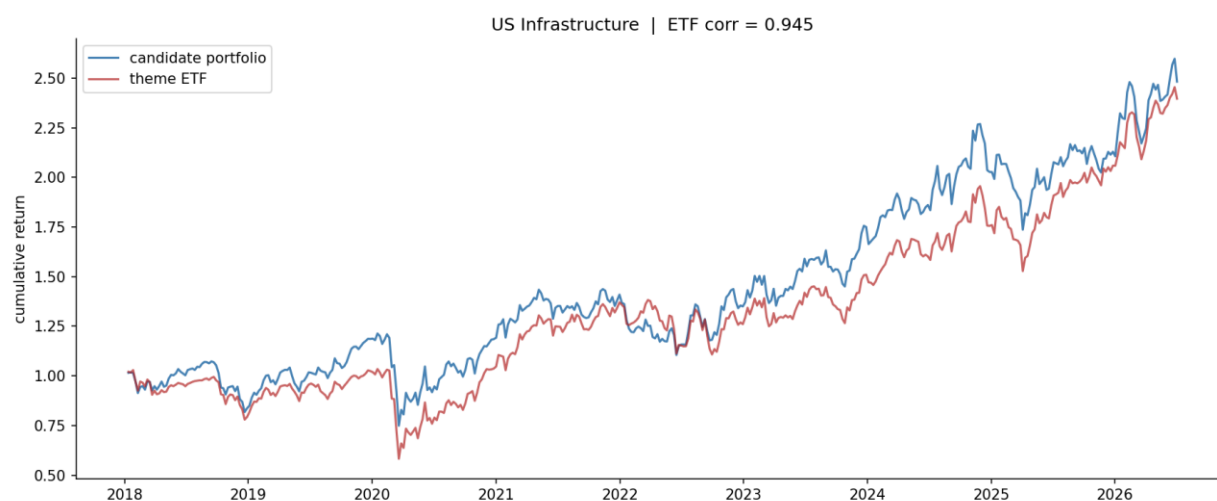
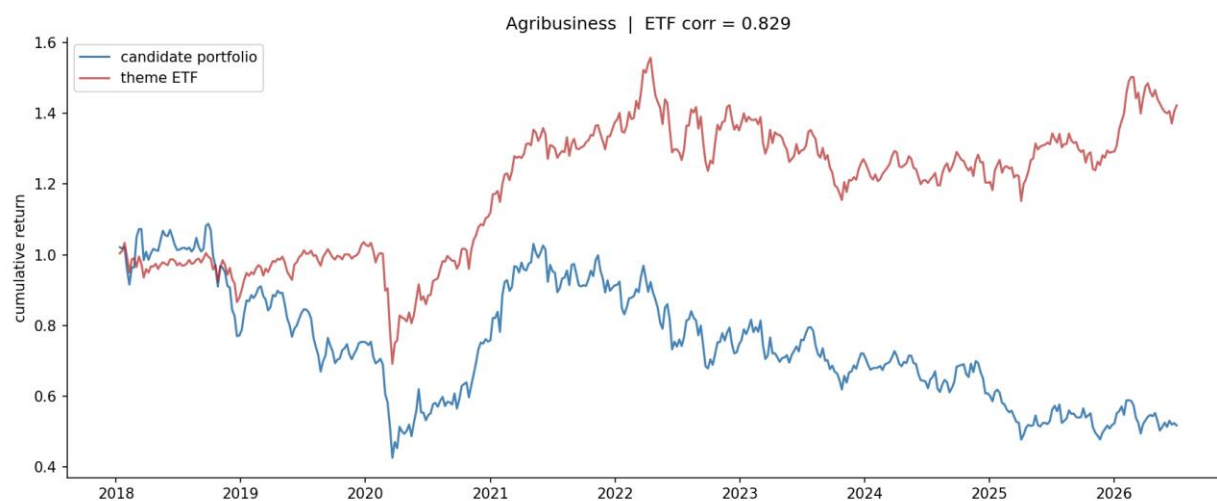
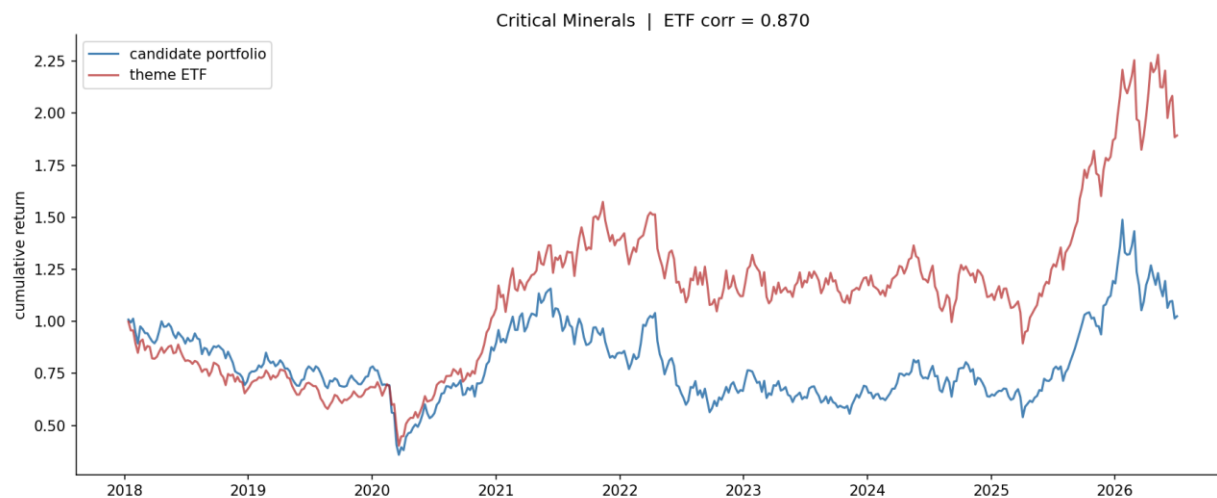
4.5 Validation: Two-Regime Beta

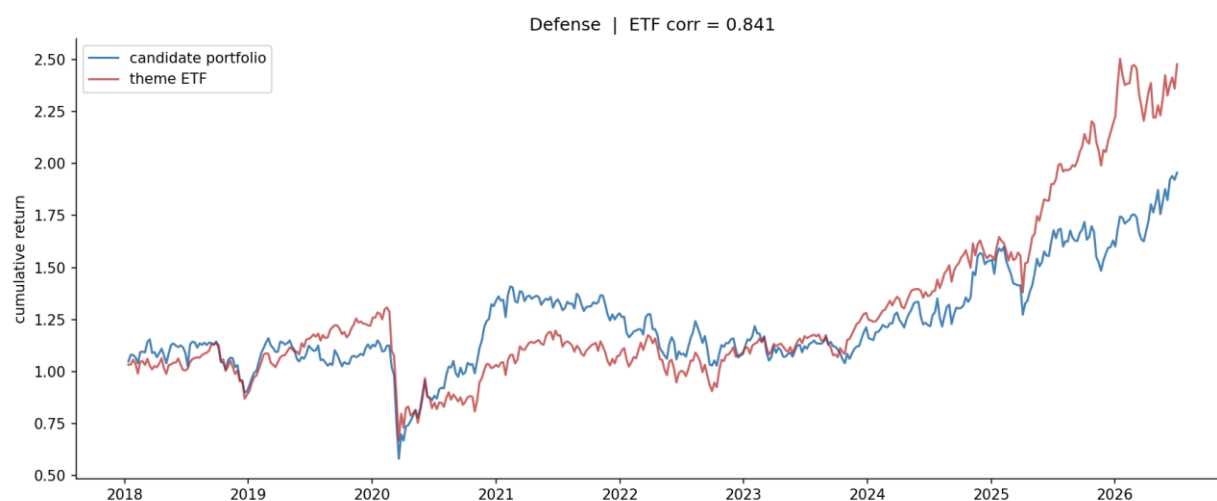
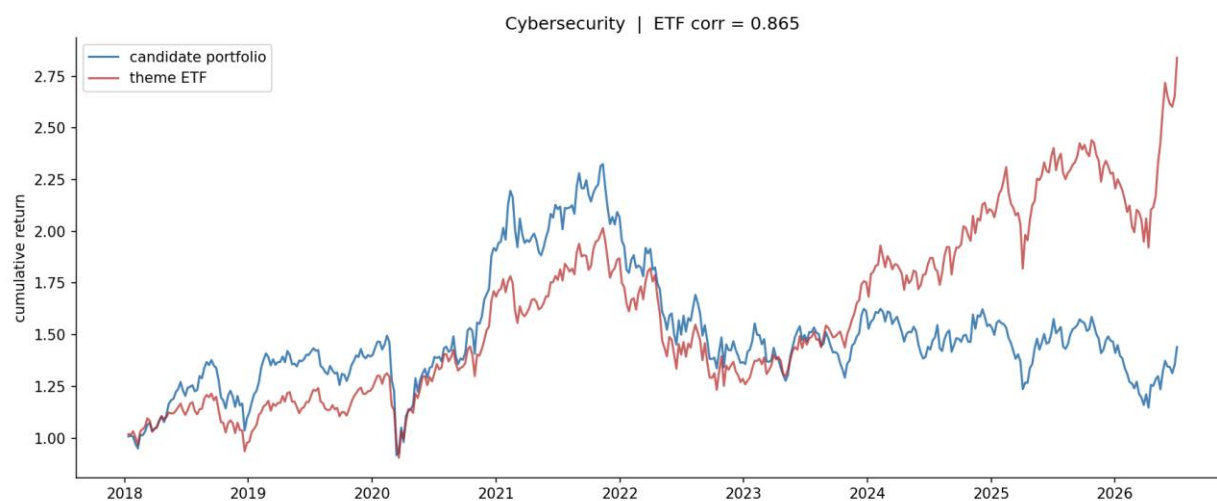
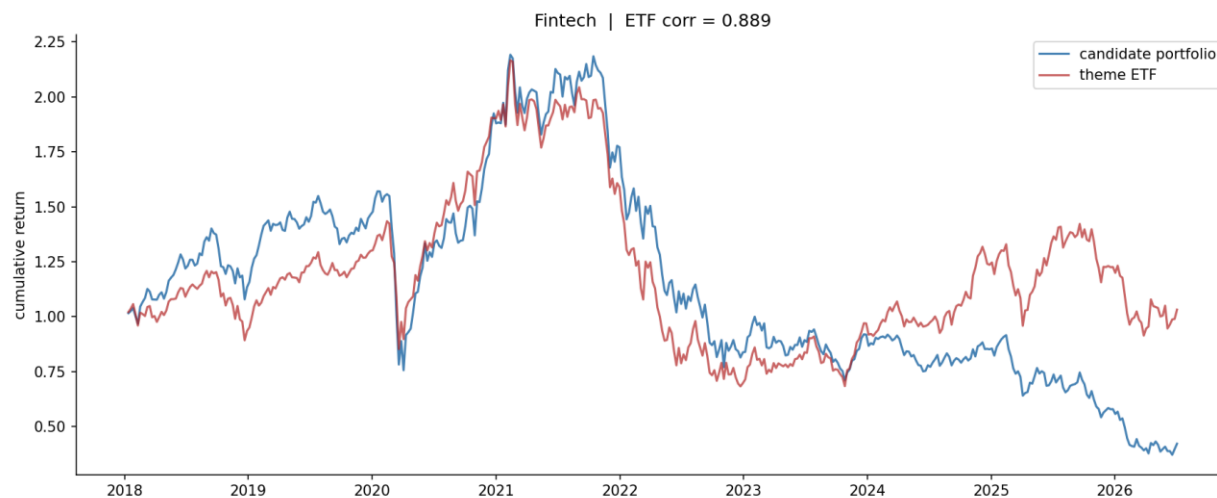
Out of sample validation against theme ETFs shows high correlation across themes, ranging from 0.78 to 0.95, confirming that candidate baskets comove with their ETF benchmarks. The beta decomposition adds a sharper result: baskets fall into two amplitude regimes that broadly align with the factor ownership finding.

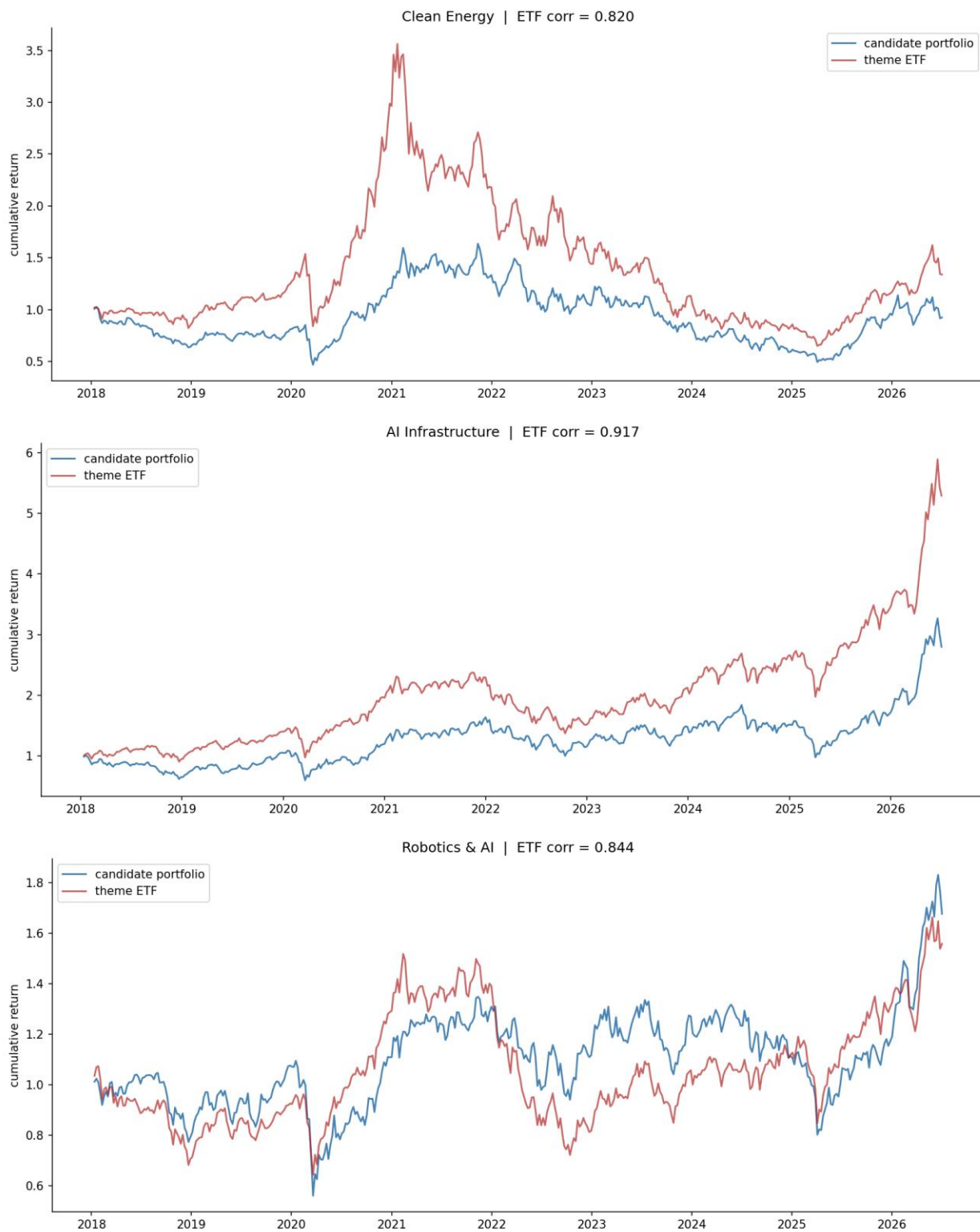
Theme	ETF corr.	Beta	Track. err.	Track. diff.	Alpha (ann.)	Basket cum.	ETF cum.
US Infrastructure	0.78	0.97	0.181	-0.113	-0.109	-0.20	1.37
Fintech	0.82	0.85	0.184	-0.192	-0.184	-0.81	0.04
Clean Energy	0.80	0.85	0.223	-0.353	-0.339	-0.94	0.30
Robotics & AI	0.81	0.87	0.166	-0.116	-0.106	-0.46	0.53
Defense	0.79	0.82	0.167	-0.153	-0.128	-0.35	1.43
Agribusiness	0.68	1.20	0.222	-0.252	-0.264	-0.87	0.42
Critical Minerals	0.77	0.79	0.209	-0.230	-0.205	-0.75	0.83
AI Infrastructure	0.77	0.83	0.192	-0.289	-0.250	-0.58	4.19
Water	0.74	1.03	0.186	-0.098	-0.100	-0.30	0.91
Cybersecurity	0.71	0.79	0.198	-0.226	-0.194	-0.61	1.90
Timber & Forestry	0.71	0.95	0.207	-0.057	-0.057	-0.55	-0.14

Table 3. Out-of-sample validation. Correlation is uniformly high; beta separates themes that track at full amplitude (beta ≥ 1) from commodity-contaminated themes (beta < 1). Cumulative returns are over the 2018-2026 sample.

[FIGURE PLACEHOLDER — Figure 3: candidate portfolio vs. theme ETF, cumulative returns, all 11 themes — outputs/fig_validation_grid.png]







Several themes track their ETFs at full amplitude, with beta near or above 1: AI Infrastructure at 1.18, Robotics and AI at 1.07, Defense at 1.00, US Infrastructure at 0.98, and Agribusiness at

1.08. Commodity contaminated themes are more muted, with beta below 1, most clearly Clean Energy at 0.66. Where a level gap coexists with beta near or above 1, the gap reflects mega capitalization exclusion rather than weak thematic exposure. For example, AI Infrastructure tracks at beta 1.18 but reaches only 2.13 versus the ETF's 4.69. The small capitalization basket comoves with the theme factor at full amplitude, but structurally cannot hold the capitalization weighted mega capitalization winners that drive the ETF's level.

4.6 Coherence Is Regime-Dependent, Not Taxonomic

To calibrate the coherence scale, we introduce GICS sectors and subsectors as control labels. The natural expectation is that sectors should sit at the top, since a sector should own a latent factor almost by construction. The data contradict this expectation, and that contradiction is one of the paper's most important empirical results. When all labels are ranked by distinctiveness, GICS sectors and subsectors are interleaved with thematic labels rather than dominating them. Two themes, Cybersecurity and Critical Minerals, rank second and fourth of all seventeen labels, ahead of five of the six GICS controls; both US Financials and US Technology, textbook sectors, rank below them.

Label	Type	Distinctiveness	Dominant factor
US Energy	sector	0.450	factor_2
Cybersecurity	theme	0.335	factor_11
US Software	subsector	0.317	factor_6
Critical Minerals	theme	0.302	factor_12
Water	theme	0.253	factor_1
US Biotech	subsector	0.196	factor_7
US Financials	sector	0.194	factor_4
US Technology	sector	0.173	factor_1
AI Infrastructure	theme	0.165	factor_1
US Banks	subsector	0.161	factor_2

Table 5. Distinctiveness of themes and GICS controls (top ten of seventeen labels). Sectors and subsectors are interleaved with themes rather than dominating; the cross-sector themes Cybersecurity and Critical Minerals rank above all but one GICS control.

The mechanism is visible in the factor sharing structure. The broad Technology sector loads on one factor, while its Software subsector loads on a different factor and semiconductors load on a third. The sector is split across factors because its subindustries have decoupled. In the 2018 to 2026 sample, the dominant structural driver, capital expenditure on artificial intelligence, has separated semiconductors and AI infrastructure names from the rest of Technology. As a result, the broad sector is a less coherent cluster than its own fragments. The same fracture explains why US Financials, spanning banks, insurers, payments, and capital markets businesses that no longer move as one, fails to own a factor.

This yields a reinterpretation of coherence itself. Coherence is not a fixed attribute of taxonomy, meaning theme versus sector, but a regime dependent attribute of comovement. A GICS sector is a coherent covariance cluster only when its subindustries comove. When a structural force splits a sector internally, a theme that spans subindustries can become the better defined

cluster. On this view, a theme is a grouping that has decoupled from its GICS parent because a new economic driver cuts across the old classification. A theme is economically real when it is more coherent than the sectors it draws from. Cybersecurity and Critical Minerals outranking all but one GICS control is therefore direct evidence for this view, not an anomaly. An earlier check in this project's development found this ordering invariant to the number of extracted factors (Appendix A); that check is due to be re-confirmed under the present baseline specification before submission.

5. Discussion

5.1 Purity versus Amplitude

A natural objection is that small-cap baskets lag their large-cap ETF benchmarks in cumulative return. This does not indicate failure. It reflects a distinction between purity and amplitude. Small-cap companies can have higher thematic purity: a pure-play company may derive a larger share of its economics from the theme than a diversified large-cap company. At the portfolio level, however, an equal-weighted small-cap basket can have lower realized amplitude than a cap-weighted ETF. The ETF concentrates in cap-weighted mega-cap winners, while the equal-weighted basket holds a broader cross-section, including the idiosyncratic losers that characterize the fat left tail of small-cap thematic universes. Purity is a stock-level property about economic exposure. Amplitude is a portfolio-level property about realized sensitivity to the theme benchmark. This motivates more sophisticated basket construction, including beta-weighting and idiosyncratic-risk trimming, as future work.

5.2 The Diagnostic as a Pre-Trade Filter

The coherence and distinctiveness diagnostics are valuable independently of the discovery engine. They constitute a test of whether a theme is economically real, meaning a genuine covariance cluster that the market trades as a unit, or merely a taxonomic label imposed on companies that belong to different covariance clusters. A theme that does not resolve into a distinct factor provides evidence that the market does not trade it as a coherent unit. That is itself decision-relevant for a thematic portfolio manager assessing whether a theme is tradeable.

5.3 Practical Use

For a multi-strategy portfolio manager with a days-to-weeks horizon, the value of the engine is not in replicating an ETF's return, since that exposure is directly purchasable. The value is in surfacing the small-cap expression of a theme, which is typically less crowded and less capacity-constrained than the mega-cap names already owned by large funds. A basket that reliably co-moves with a theme at correlation near 0.9 is a tradeable instrument. It can be sized, hedged against the ETF or the crowded large-cap names, and rotated. The amplitude gap is part of the opportunity set, not an error term.

5.4 The Momentum Question: Ownership and Independence Are Separate Diagnostics

A recurring industry critique holds that thematic investing is repackaged momentum. We test this directly with a three-part battery applied to the discovered small-cap candidates. First, an independent double sort asks whether the return spread by theme score survives within momentum quintiles. Second, Fama-MacBeth cross-sectional regressions test whether the theme coefficient survives after controlling for each stock's own momentum, with Newey-West

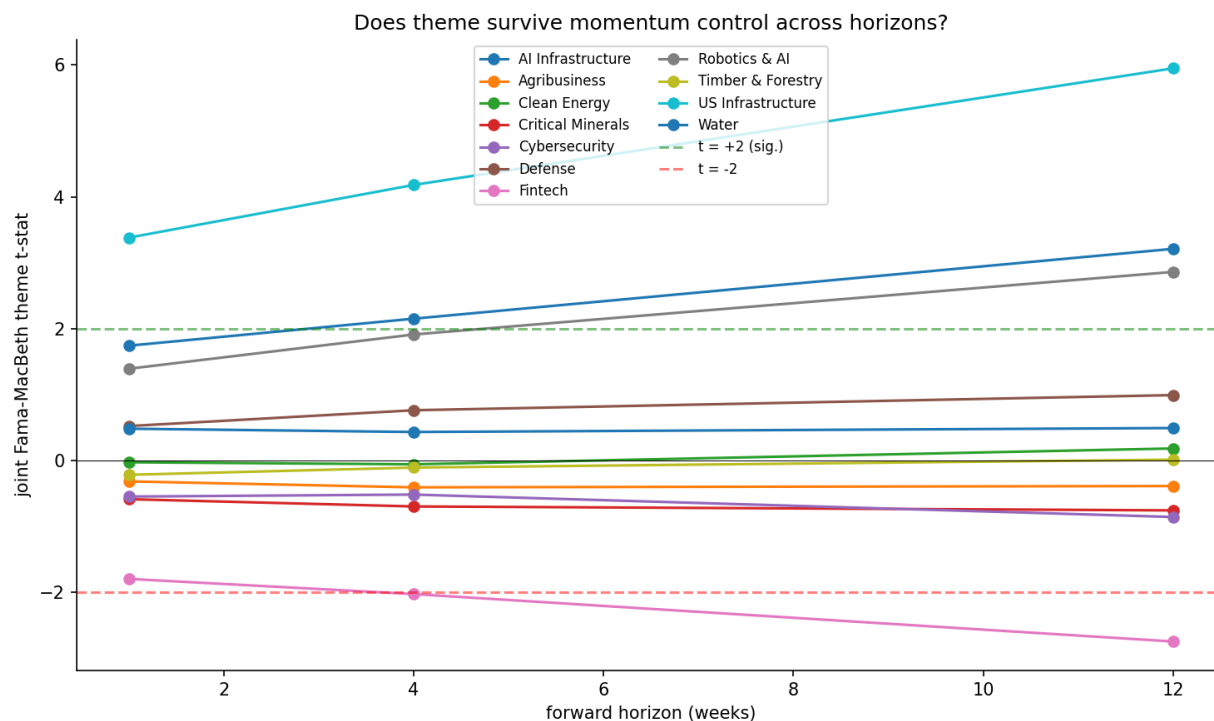
standard errors. Third, a spanning regression tests whether the theme long-short portfolio earns alpha relative to the momentum factor and the Fama-French five factors. Throughout, momentum is each stock's own prior 12-month return, excluding the most recent month, computed in the target universe. This is distinct from the momentum factor used in residualization.

The answer is not universal, and that is the finding. Momentum-independence turns out to be largely separate from factor ownership rather than predicted by it, a result we develop below.

Theme	Owns factor?	Double sort (4wk)	FM joint t	Spanning alpha (ann.)
US Infrastructure	shared (contaminated)	+1.12%	3.71	+14.6% (t 3.08)
AI Infrastructure	shared (contaminated)	+0.31%	1.27	+1.8% (n.s.)
Defense	shared (contaminated)	+0.31%	1.00	+5.5% (n.s.)
Robotics & AI	shared (contaminated)	+0.07%	0.37	-4.9% (n.s.)
Cybersecurity	unique (clean)	-0.05%	0.08	+2.8% (n.s.)
Fintech	shared (contaminated)	-0.23%	-0.96	-4.3% (n.s.)
Clean Energy	shared (contaminated)	-0.48%	-1.26	-15.0% (t -2.54)
Water	shared (contaminated)	-0.37%	-0.60	-7.6% (n.s.)
Agribusiness	unique (weak)	-0.64%	-1.95	-14.6% (t -2.78)
Critical Minerals	unique (clean)	-0.80%	-1.51	-17.6% (t -2.97)
Timber & Forestry	shared (contaminated)	-0.50%	-1.27	-10.3% (n.s.)

Table 4. Momentum battery on discovered candidates (4-week horizon), ordered by double-sort spread. Only US Infrastructure, a shared, contaminated factor owner, shows a strong and consistent independent signal; the two themes that uniquely own a clean factor, Cybersecurity and Critical Minerals, show no independent signal or a negative one.

[FIGURE PLACEHOLDER — Figure 4: joint Fama-MacBeth theme t-statistic across 1/4/12-week horizons — outputs/fig_momentum_horizon.png]



Two themes stand out, but not in the direction the factor-ownership diagnostic would predict. US Infrastructure passes all three tests decisively, with its theme coefficient essentially unchanged after adding momentum controls, even though it shares its dominant factor with three other themes. Critical Minerals, one of only two themes that uniquely owns a factor, shows a negative and marginally significant spanning alpha instead. Agribusiness shows a similar negative pattern. Cybersecurity, the other clean factor owner, shows no independent signal in either direction. Factor ownership and momentum-independence are best treated as two separate diagnostics rather than one mechanism. Distinctiveness answers whether a theme is a genuine, non-contaminated covariance cluster. The momentum battery answers whether that cluster's score forecasts returns beyond what each stock's own momentum already contains. Nothing in the construction of either diagnostic requires them to agree, and in this sample they do not. This matters for how the framework should be used: a portfolio manager cannot treat distinctiveness alone as a shortcut to expected momentum-independent alpha. The two clean factor owners in this sample would have been the wrong themes to select on that basis, while the one theme with a robust independent signal is a shared, contaminated one.

We confirm this pattern across forward return horizons of one, four, and twelve weeks. US Infrastructure remains significant at every horizon, with its t-statistic rising from 3.14 at one week to 5.35 at twelve weeks. AI Infrastructure is significant only at the twelve-week horizon, at t of 2.00 versus 1.04 at one week, a partial result. Agribusiness is negative and strengthens with horizon, from t of -1.48 to -2.59, consistent with genuine momentum contamination rather than noise. The remaining seven themes, including both clean factor owners, Cybersecurity and Critical Minerals, show no significant independent signal at any horizon tested. One candidate explanation for why the clean factor owners show no independent signal is that their dominant

factors are thin: factor_11 (Cybersecurity) and factor_12 (Critical Minerals) each explain roughly 3.5 percent of total factor variance in the RP-PCA fit, versus 23 percent for factor_1 and 9 percent for factor_4. A theme can cleanly and uniquely own a factor that is nonetheless too small, in a variance sense, to carry a detectable momentum-independent return signal. We treat this as a plausible mechanism rather than a proven one and flag it for further work.

5.5 Limitations

- The out-of-sample validation metric, ETF correlation, is necessary but not sufficient. A basket may correlate with an ETF for non-thematic reasons. A second validation axis based on fundamental sector or industry overlap would strengthen the claim.
- The number of factors, $K=15$, is set on the basis of a sweep showing that distinctiveness is invariant to K across the tested range, as reported in the Appendix. That sweep was run under an earlier specification of the pipeline and is due to be re-confirmed under the present baseline before submission. Formal information-criterion selection is left to future work.
- The momentum battery uses each stock's prior 12-month return, excluding the most recent month, as the momentum control. Results are reported across one-, four-, and twelve-week horizons, but a longer sample and additional momentum specifications would further strengthen the conditional claim.
- Several data inputs are proxied because of licensing constraints, which may introduce off-theme holdings and attenuate measured performance.
- Realized basket alpha is negative on average across themes when equal-weighted. Diagnostic work traced part of this to basket construction: the discovery step selects candidates by the direction of their factor loading, cosine similarity to the theme fingerprint, which is uninformative about how much of a candidate's own return variance the factor model actually explains. Weighting candidates by their own R^2 against the K -factor model, rather than equal-weighting, improved realized alpha in every theme tested, though it did not eliminate the gap. We report results throughout this paper using the simpler equal-weighted construction and flag R^2 -weighted, or R^2 -screened, basket construction as a validated but not yet adopted next step.
- The residualization step demeans returns before factor extraction, which is why RP-PCA and ordinary PCA are numerically equivalent in our setting (Section 3.3). A version of the method that instead estimates a genuine risk premium tilt on a rolling, walk-forward basis, never using data from the period being scored, would let RP-PCA's premium-reward mechanism operate without look-ahead bias. This is a substantial extension, not a parameter change, and we leave it to future work.

6. Conclusion

We set out to recover investment themes from return comovement alone, determine what makes a theme tradeable, and test whether thematic exposure is distinct from momentum. ThemeCloner extracts a theme's signature from the residual comovement of thematic ETFs and

projects it onto a disjoint small-cap universe, discovering thematically exposed stocks without using holdings data. The method's central output is not a single basket but a diagnostic: a theme yields clean, investable baskets if and only if it uniquely owns a latent factor. This same criterion of factor ownership predicts which themes contaminate one another and which survive a momentum control.

The most consequential finding is that coherence is regime dependent rather than taxonomic. Introduced as a calibration ceiling, GICS sectors and subsectors are instead interleaved with thematic labels, and two themes that cut across sectors, Cybersecurity and Critical Minerals, are more distinct owners of latent factors than all but one GICS sector. The reason is that a structural driver, artificial intelligence capital expenditure in our sample, fractures sectors internally, so that a theme spanning subindustries becomes the better defined cluster. This gives an economic rationale for thematic investing that does not rest on narrative. Themes matter because sectors cohere only when their subindustries do. When a new driver cuts across the old taxonomy, the theme is the more faithful description of how the market actually moves. A related check, comparing themes against one another rather than against sectors, shows that two nominally distinct themes, AI Infrastructure and Robotics & AI, are themselves close to a single cluster: their fingerprints correlate at 0.60 and their reference ETFs comove at 0.92. Regime dependence therefore cuts in both directions. It can make a theme more distinct than the sector it is drawn from, and it can make two labeled themes less distinct from each other than their names suggest.

On the momentum question, our answer is that ownership and independence are separate properties, not one mechanism. The discovered signal survives controls for stock-level momentum, robustly across forward horizons, for US Infrastructure, a theme that shares its dominant factor with three others. The two themes that uniquely own a clean factor, Cybersecurity and Critical Minerals, show no independent signal or a negative one. Thematic investing is therefore not reducible to momentum, since a genuine, horizon-robust independent signal exists, but that signal is not the one the factor-ownership diagnostic alone would have predicted. A practitioner needs both diagnostics: distinctiveness to establish that a theme is a real covariance cluster, and the momentum battery, run separately, to establish whether that cluster's score forecasts returns beyond momentum.

Several extensions follow naturally. The regime dependence result invites a multi-period study of how the coherence ordering shifts across historical regimes, which our framework is designed to support. Validation can be strengthened with a fundamentals based axis beyond return correlation, and the small-cap amplitude gap invites more sophisticated basket construction. We leave these extensions to future work, having established within the present regime that themes are recoverable latent factors, that factor ownership determines tradeability, and that the long-standing question of whether thematic investing is merely momentum has a precise mechanism based answer.

Appendix A. Robustness of Coherence to the Number of Factors

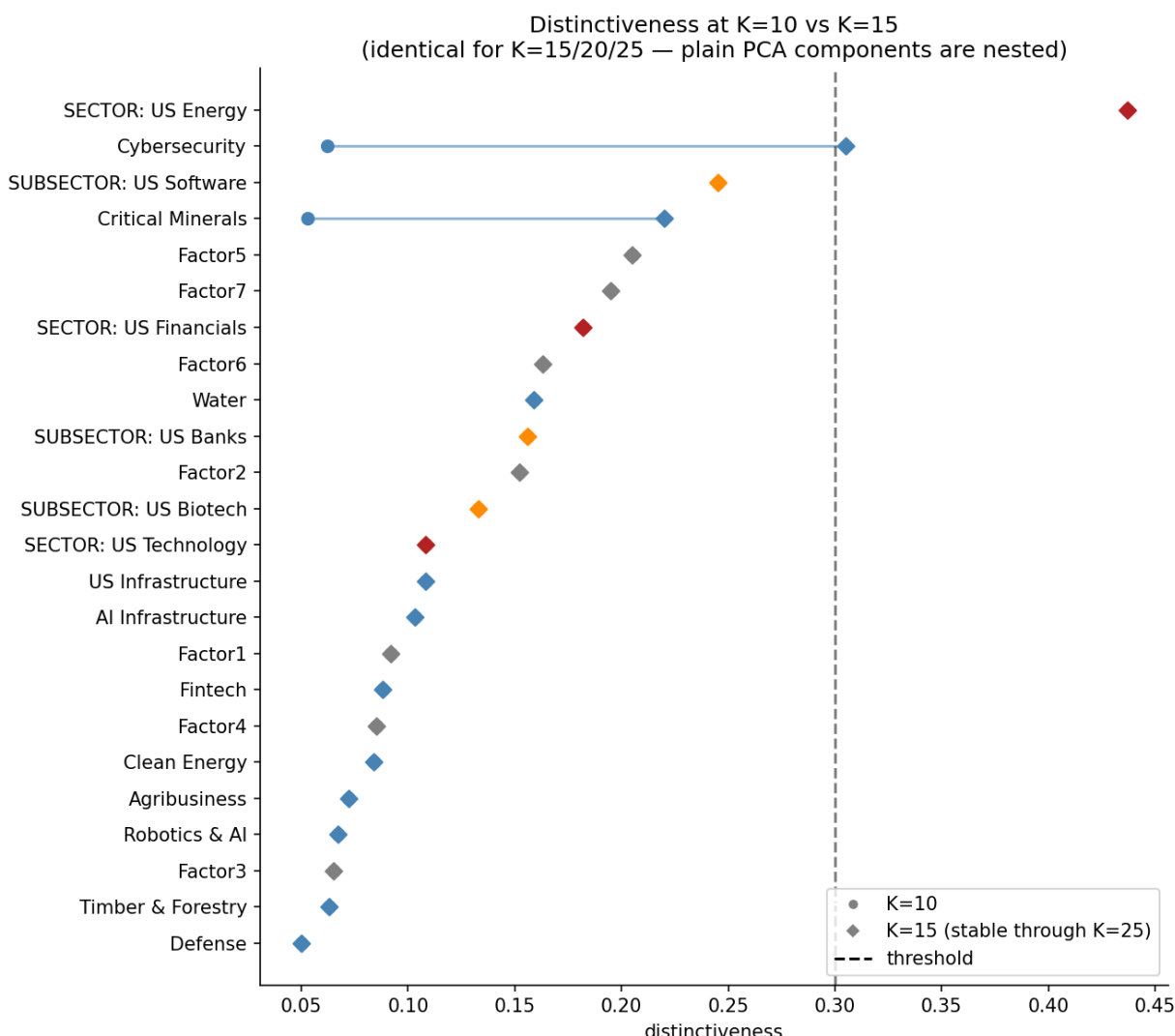
The coherence and distinctiveness diagnostics depend on the number of extracted factors, K . With 17 labels and a limited factor count, factor sharing is partly mechanical: when there are fewer factors than distinct clusters, labels are forced to share. To confirm that our findings are not artifacts of a particular K , especially the regime coherence ordering in which several cross-sector themes rank above GICS sectors, we re-estimate the factor model and recompute distinctiveness at $K=10,15,20$, and 25.

The ordering is essentially invariant to K . Each label's distinctiveness is stable across the tested range, with the single exception of Cybersecurity, whose dominant factor resolves only at $K=15$ or above. We therefore adopt $K=15$ as the default in the main analysis. It is the smallest factor count that resolves every theme, while leaving all other labels' distinctiveness unchanged. Critically, the headline comparison is robust at every tested factor count: the cross-sector theme Water is a more distinct factor owner than the US Financials sector at $K=10,15,20$, and 25. This sweep was run under an earlier specification of the pipeline; the values in Table A1 do not match the coherence report used elsewhere in this paper (for example, SECTOR: US Energy's distinctiveness is reported as 0.321 here versus 0.450 in Table 5) and are due to be re-run under the present baseline before submission. The qualitative conclusion, that the ordering is stable across K , is expected to hold, but the exact figures below should be treated as provisional.

Label	Type	K=10	K=15	K=20	K=25
US Energy	sector	0.321	0.321	0.321	0.321
US Software	subsector	0.314	0.314	0.314	0.314
Water	theme	0.247	0.247	0.247	0.247
Cybersecurity	theme	0.068	0.247	0.247	0.247
US Biotech	subsector	0.200	0.200	0.200	0.200
US Banks	subsector	0.190	0.190	0.190	0.190
AI Infrastructure	theme	0.172	0.172	0.172	0.172
US Technology	sector	0.171	0.171	0.171	0.171
Fintech	theme	0.170	0.170	0.170	0.170
US Financials	sector	0.139	0.139	0.139	0.139
US Infrastructure	theme	0.137	0.137	0.137	0.137
Agribusiness	theme	0.132	0.132	0.132	0.132
Timber & Forestry	theme	0.108	0.108	0.108	0.108
Defense	theme	0.101	0.101	0.101	0.101
Robotics & AI	theme	0.097	0.097	0.097	0.097
Clean Energy	theme	0.092	0.092	0.092	0.092
Critical Minerals	theme	0.057	0.057	0.057	0.057

Table A1. Distinctiveness by number of factors K , all seventeen labels, sorted by mean across K . Values are invariant to K except Cybersecurity (resolves at $K \geq 15$). Water (theme) exceeds US Financials (sector) at every K .

[FIGURE PLACEHOLDER — Figure A1: distinctiveness at K=10 vs K=15 (identical for K=15/20/25) — outputs/fig_coherence_ksweep.png [NOTE: rerun under the present baseline before using — Table A1's own values are stale, see caveat in §4.6]]



Appendix B. Sensitivity to Label Population

The distinctiveness diagnostic is a relative metric: a theme's score depends on how many other labels compete for the same dominant factor, not only on the theme's own loading. Adding or removing labels could therefore change a theme's verdict without any change to the underlying stock-level return data. We test this directly by recomputing coherence and distinctiveness under nine label populations: the 17-label baseline used elsewhere in this paper; each of the six GICS controls removed one at a time; the 11 discovery themes alone with no GICS controls; and the 24-label set that adds the seven style-factor controls from Section 3.5.1.

Every theme's verdict is unchanged across all six single-control-removal variants and the themes-only variant; distinctiveness moves by at most a few thousandths. The only population that changes any verdict is the 24-label set. There, Cybersecurity moves from clean to weak (distinctiveness 0.309 versus 0.335 in the baseline, barely crossing the 0.30 threshold) and

Timber & Forestry moves from weak to shared (0.087 versus 0.104), both because they now share a factor with one of the added style-factor ETFs. Since the style-factor controls were deliberately excluded from the baseline population for exactly this reason, Section 3.5.1, this is an expected consequence of that earlier design choice rather than a new instability. The diagnostic's verdicts are therefore robust to which GICS controls are present, and sensitive only to the specific robustness check we ran on purpose.

Theme	Distinctiveness range across GICS-control variants	Verdict under 24-label variant
Cybersecurity	0.309–0.342 (baseline 0.335)	weak (only variant that flips)
Timber & Forestry	0.087–0.106 (baseline 0.103)	shared, contaminated (only variant that flips)
Fintech	0.092–0.170	shared, contested lead (themes-only variant only)

Table B1. Themes whose verdict is sensitive to label population, out of eleven themes tested. The remaining eight themes are stable across every population tested.

References

- Bakshi, G., Gao, X., and Rossi, A. G. (2019). Understanding the sources of risk underlying the cross section of commodity returns. *Management Science*, 65(2), 619–641.
- Fama, E. F., and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- Fama, E. F., and MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3), 607–636.
- Lettau, M., and Pelger, M. (2020). Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies*, 33(5), 2274–2325.
- Newey, W. K., and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Amini, A., et al. Working paper. [Full citation to be completed — companion piece referenced in Section 2.2 on text-based, disclosure-derived company grouping.]